Exploring the *data.*

EDA & data preprocessing for demand forecasting in an AI-native supply chain.

DATASET
Telecom Market in Bangladesh

PERIOD
Jan 2008 – Dec 2015 · 96 months

ENGINE
ARIMA · LSTM · XGBoost ensemble

THEN
3 lot-sizing algorithms

GROUP · DBI MAJOR

	COURSE PROJECT
Muhammad Kawser Ibn Khurshid Khan	24224525
Asir Abdullah Chowdhury	24224511
Labib Mashfiq Rahman	23229564
Islam MD Mujahidul	23229550

Course · Business Analytics and AI Project

01 / WHAT IS CHRONOFORGE PULSE?

An AI-native engine for *forecasting & lot-sizing*.

A decision platform for supply-chain planning. Inputs are demand signals — telecom subscribers; outputs are forecasts plus buy plans. **This presentation prepares the data so the engine can learn from it.**

ENSEMBLE OF THREE MODEL FAMILIES

ARIMA	Classical statistical model — captures seasonality and trend.
LSTM	Deep learning — long-range temporal dependencies.
XGBoost	Gradient-boosted trees — non-linear interactions.
+ Lot-sizing	Three algorithms turn forecasts into order quantities.

INPUTS Demand signals · e.g., telecom subscribers

OUTPUTS Forecasts + buy plans · feeds ARIMA · LSTM · XGB + 3 lot-sizers

02 / THREE PRESENTATIONS, ONE PLATFORM

Where *today* fits.

01

PRESENTATION 1

Problem Framing & Business Context

Define ChronoForge Pulse, the use case, and the data source.

→

TODAY

02

PRESENTATION 2

EDA & Data Preprocessing

Clean, reshape, profile, and engineer features for the model.

→

03

PRESENTATION 3

ML Modelling & Lot-Sizing

Train ARIMA, LSTM, XGBoost; run three lot-sizing algorithms.

03 / WHAT WE DO TODAY VS. WHAT COMES NEXT

From raw CSV to *model-ready*.

In Scope TODAY

- Load raw telecom CSV `header=None`
- Reshape wide → long (tidy)
- Type casting & date parsing
- Missing-value & duplicate checks
- Feature engineering — `MoM_Growth_%`, `Market_Share_%`
- Descriptive statistics & visual EDA (5 charts)
- Outlier detection (IQR) & export model-ready dataset

Next PRESENTATION 3

- Train ARIMA, LSTM, XGBoost forecasts
- Build the ensemble blend
- Evaluate — MAPE / RMSE / MAE
- Apply 3 lot-sizing algorithms — EOQ, Wagner-Whitin, Silver-Meal
- Generate order plans & compare scenarios
- Deploy ChronoForge Pulse

04 / WHAT THE PREPROCESSING BLOCK TAKES AND PRODUCES

Wide & messy *in*. Tidy & tabular *out*.

INPUT

78 × 112

`GP Model.xlsx — Telecom Mkt. in Bangladesh.csv`

RAW FEATURES

Operator — 6: GP, Banglalink, Robi, Airtel, Citycell, Teletalk. **Year** 2008–2015, **Month** Jan–Dec, **Subs** in millions across 96 columns.

DROPPED

Sector-total row · Avg-12M / assumption columns · blank padding rows & cols. Granularity: monthly.

EDA + PREPROCESSING

10 steps

Load raw CSV

Slice subscriber block

Reshape wide → long (melt)

Parse Date, Year, Month

Coerce numerics; check NaN

Drop duplicates

Engineer features

Compute stats & IQR outliers

Visualise (line / box / bar / heat)

Save telecom_clean.csv

OUTPUT

576 × 7

`telecom_clean.csv`

COLUMNS

Operator · Date `(datetime64)` · Year · Month · Subscribers_Mn · MoM_Growth_% · Market_Share_%

IQR outliers flagged	**51**
Nulls / duplicates	**0 / 0**
Ready for	**ARIMA · LSTM · XGB**

HEADER NOTE

Multi-row header: Year row stacked above Month row · coerced into a single Period string before .melt()

05 / TELECOM MARKET IN BANGLADESH

Eight years of *subscriber demand*.

SOURCE	Grameenphone internal market model (based on BTRC data)
PERIOD	January 2008 – December 2015 · 96 months
OPERATORS	Grameenphone · Banglalink · Robi · Airtel · Citycell · Teletalk
UNITS	Subscribers in millions, one value per operator per month

RAW LAYOUT

78×112

wide · mixed with assumption rows

AFTER CLEANING

576×7

one row per Operator × Month

"Why telecom data?"

Subscribers are a strong proxy for downstream demand — SIM cards, handsets, recharge logistics, tower equipment.

SECTOR GROWTH **506 → 1,536 Mn-months · ~3× over 8 years**

PROXY FOR SIM cards · handsets · recharge · tower equipment

06 / WIDE FORMAT — MONTHS AS COLUMNS

A spreadsheet, *not a table*.

RAW LAYOUT (SIMPLIFIED) — VALUES IN MILLIONS

OPERATOR	JAN-08	FEB-08	…	DEC-15
Grameenphone	16.88	17.20	…	56.54
Banglalink	7.45	7.88	…	33.39
Robi	7.28	7.36	…	28.94
Airtel	2.37	2.60	…	9.82
Citycell	1.45	1.51	…	1.05
Teletalk	1.00	1.00	…	4.28

CSV loaded with `header=None` — multi-row header layout

Rows **4–10** contain operator subscriber data

Columns **2–97** hold monthly values (Jan-2008 → Dec-2015)

Subscriber values in **millions**

Extra trailing columns hold averages / assumptions — `dropped`

07 / ONE ROW PER (OPERATOR, MONTH)

Unpivot to *tidy*.

Sliced rows 4–10 and columns 2:98 into an operator × month matrix, coerced to numeric, then `pandas .melt()` unpivots into long format. **6 operators × 96 months = 576 tidy rows.**

WIDE · 78 × 112 — BEFORE

OPERATOR	JAN-08	FEB-08	…
Grameenphone	16.88	17.20	…
Banglalink	7.45	7.88	…
Robi	7.28	7.36	…
Airtel	2.37	2.60	…

.MELT() →

LONG · 576 × 7 — AFTER

OPERATOR	DATE	SUBS_MN
Grameenphone	2008-01	16.88
Grameenphone	2008-02	17.20
Banglalink	2008-01	7.45
Banglalink	2008-02	7.88

08 / MISSING VALUES, DUPLICATES, TYPES

Clean by *construction*.

576 ROWS · OPERATOR × MONTH	7 COLUMNS
0 NULLS AFTER TIDYING	0 DUPLICATE (OPERATOR, DATE) KEYS

Missing values per column: **all zero** after tidying

Safety net: `groupby-interpolate → ffill → bfill` on Subscribers_Mn

Types verified: **Date** is `datetime64`, **Subscribers_Mn** is `float64`

Numerical: Year, Subscribers_Mn, MoM_Growth_%, Market_Share_%

Categorical: Operator, Month

ROW MATH

576 rows = 6 operators × 96 months · zero loss vs. raw 78×112 source

09 / NEW VARIABLES THE MODELS WILL USE

Two features that *carry the signal.*

MoM_Growth_%

`= pct_change(Subscribers_Mn) within each Operator`

Captures monthly growth dynamics — a key input for XGBoost and LSTM.

Market_Share_%

`= Subscribers_Mn / Sector total (per Date) × 100`

Captures relative competitive position over time.

Year + Month name extracted → seasonality & calendar features

These features feed directly into Presentation 3 forecasting models

Final dataset saved as `telecom_clean.csv` **(576 × 7)**

10 / PER-OPERATOR SUBSCRIBER STATISTICS (MILLIONS)

A market with *one giant*.

OPERATOR	MIN	MEAN	MAX	STD
Grameenphone	16.88	**36.03**	56.54	12.16
Banglalink	7.45	**21.68**	33.39	8.08
Robi	7.28	**17.22**	28.94	7.20
Airtel	2.20	**5.69**	9.82	2.43
Teletalk	0.84	**1.93**	4.28	1.18
Citycell	1.05	**1.61**	2.00	0.27

Grameenphone leads on every measure — mean **36.03 Mn**, peak **56.54 Mn**

High std (12.16, 8.08, 7.20) for the top three → strong growth, not noise

Citycell is flat & tiny — std **0.27**, range barely 1 Mn

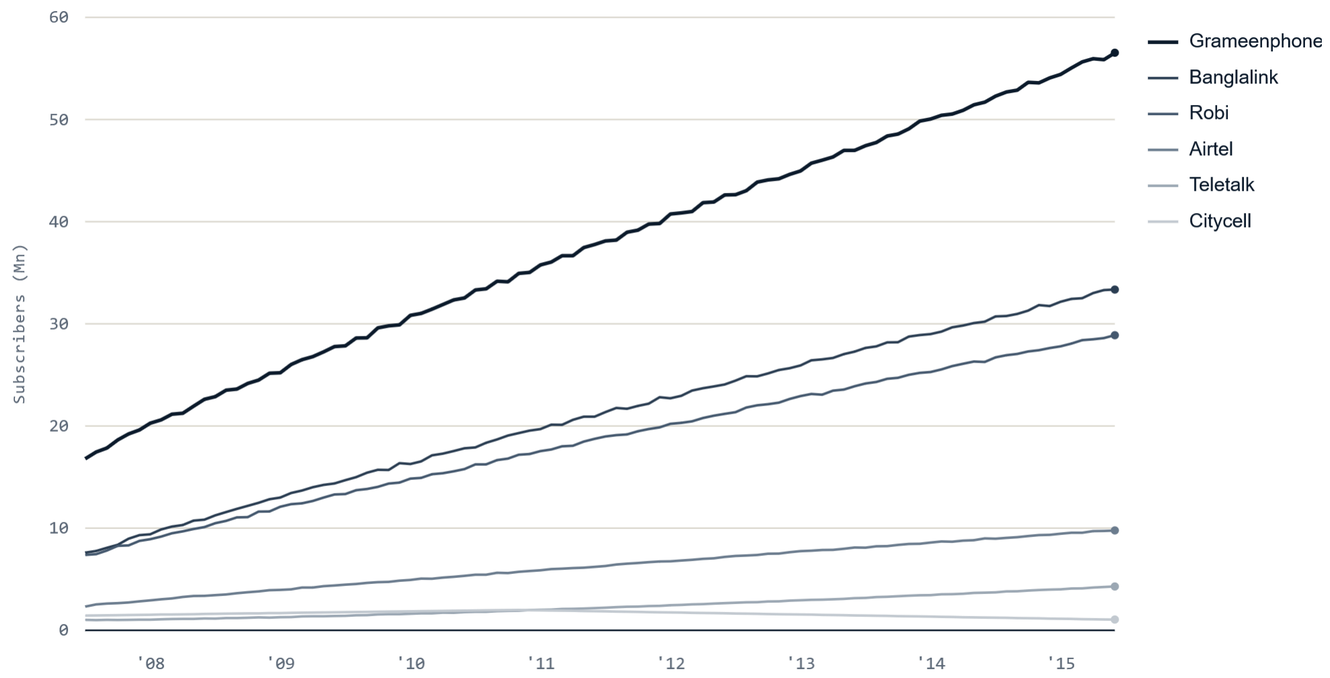
GAP *Leader-to-laggard: Grameenphone mean (36.03 Mn) ≈ 22× Citycell (1.61 Mn) · the top three (GP · Banglalink · Robi) carry ~89% of sector demand.*

11 / MONTHLY SUBSCRIBERS PER OPERATOR · 2008–2015

The whole market is *climbing*.

SUBSCRIBERS (MN) OVER TIME

LINE · 6 OPERATORS



Three clear winners.

Grameenphone, Banglalink, Robi show strong monotonic growth — ideal for trend models.

Citycell is the exception.

It rises then declines toward 2015 — a candidate for a downward forecast.

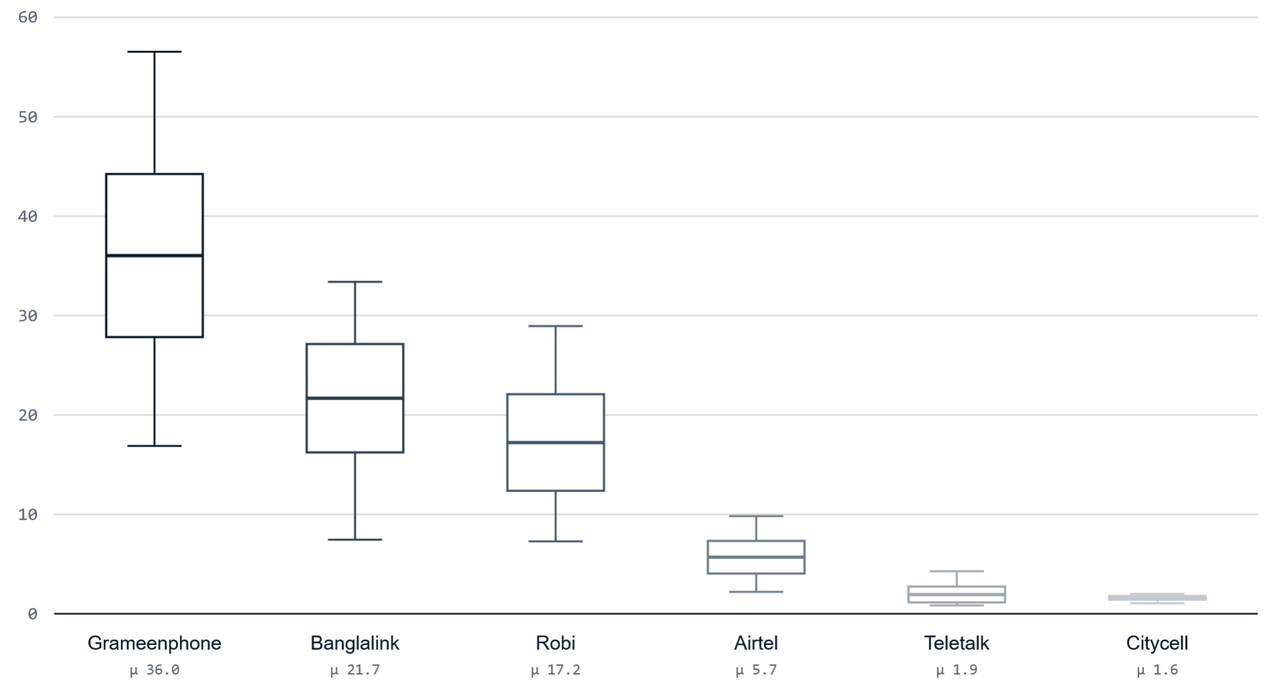
Wide dynamic range.

From sub-1 Mn to 56 Mn on one axis — the spread itself motivates an ensemble.

12 / BOXPLOT REVEALS SPREAD & RANGE

Spread tells the *growth story*.

SUBSCRIBER DISTRIBUTION (MN) BOX · PER OPERATOR



Tall boxes = big movers.

Grameenphone spans 17–57 Mn; the height is the eight-year growth made visible.

Tight boxes = stable players.

Citycell sits in a thin band near 1.6 Mn — almost no variation.

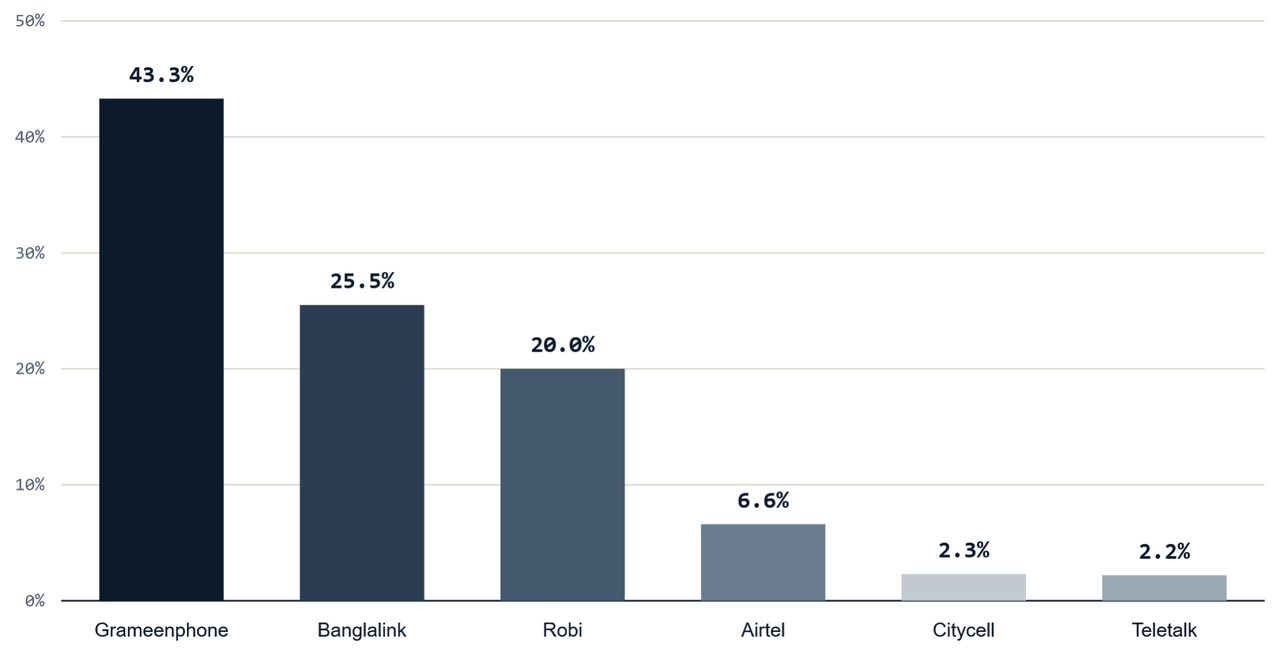
Box ≈ μ ± 0.67σ.

Whiskers mark min/max; the median line approximates each operator's mean.

13 / WHO DOMINATES THE TELECOM SECTOR?

Three operators own *89%*.

AVERAGE MARKET SHARE (%) BAR · PER OPERATOR



A clear leader.

Grameenphone at 43.3% holds nearly half the market on its own.

The top three.

GP + Banglalink (25.5%) + Robi (20.0%) = **~89%** of all subscribers.

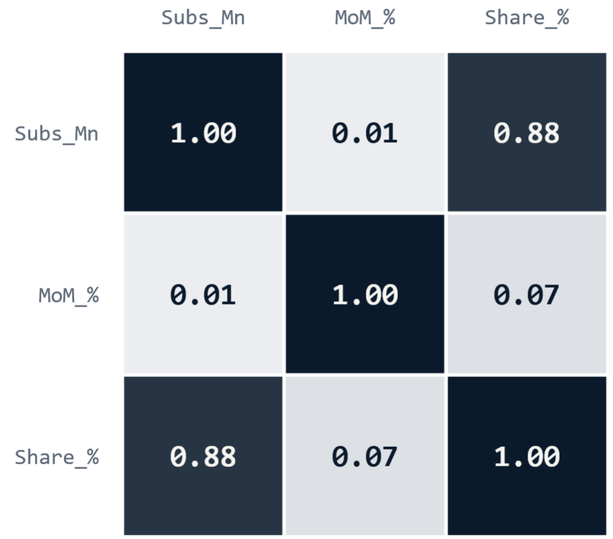
The long tail.

Airtel, Citycell, Teletalk together hold barely 11% — lumpy, low-volume demand.

14 / TWO VIEWS THAT CONFIRM THE DEMAND STORY

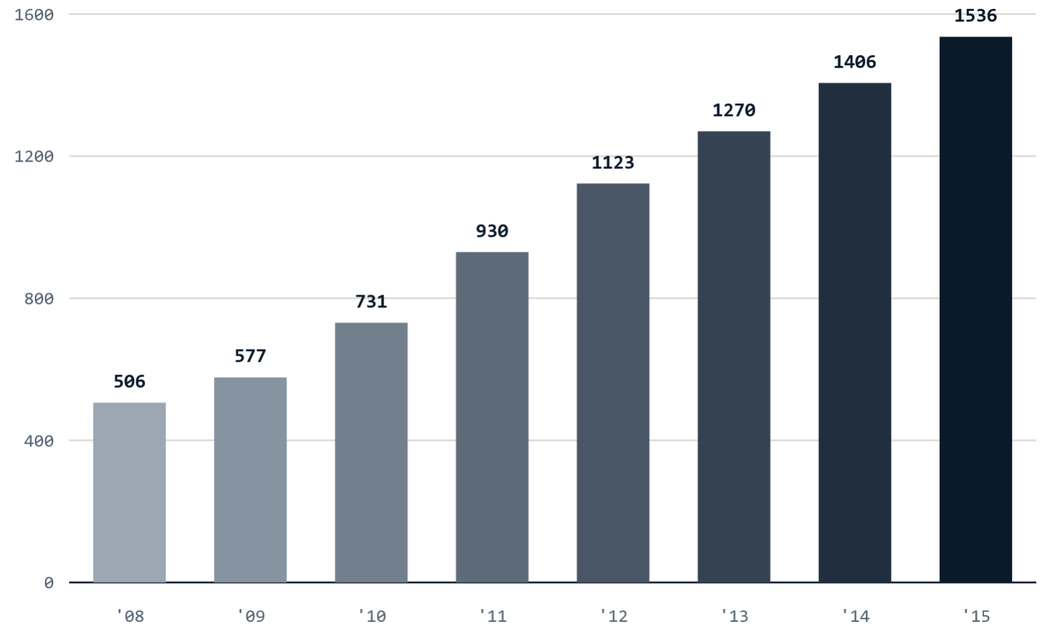
Share *is* demand. Demand is *rising*.

CORRELATION MATRIX



Subscribers ↔ Market_Share = **0.88** — forecasting share is essentially forecasting demand mix. MoM growth is near-independent of level (0.01).

YEARLY SECTOR TOTAL (MN, SUMMED MONTHLY)



Total demand grew **~3× in eight years** — 506 → 1,536 Mn-months. A strong upward trend for ARIMA & LSTM to project.

15 / DEMAND SHOCKS ANY FORECASTER MUST HANDLE

51 outliers — *real shocks*, not errors.

IQR ON MOM_GROWTH_% · Q1 − 1.5·IQR … Q3 + 1.5·IQR

Lower bound	−3.07%
Upper bound	5.50%
Observations outside the band	51

Concentrated in early-period growth spurts — **Airtel, Banglalink, Robi.** These are not data errors; they are recurring demand shocks.

- **LSTM & XGBoost** must model this volatility.
- **Lot-sizing** must keep safety stock for surge months.

CONSEQUENCE

Shocks are not noise · LSTM + XGBoost must learn the volatility · lot-sizing must hold safety stock for surge months.

16 / HYPOTHESES PRESENTATION 3 WILL QUANTIFY

From *patterns* to *predictions*.

SECTOR TREND

~3× growth in 8 years

506 → 1,536 Mn-months. **ARIMA + LSTM** should project continued upward trend into 2016+.

OPERATOR SHAPE

Three trends, one decline

GP, Banglalink, Robi → strong monotonic growth. **Citycell** declining — candidate for exit.

VOLATILITY

Shocks are recurring

51 MoM outliers → the **ensemble** lets each model cover a different regime.

MARKET-SHARE SIGNAL

Share ≈ demand

Subscribers ↔ share correlation **0.88** → forecasting share forecasts the demand mix.

LOT-SIZING

Big vs. lumpy

Top 3 (~89% share) dominate order quantities. `EOQ` for stable giants; `Silver-Meal / Wagner-Whitin` for lumpy small ops.

LOW-VOLUME

Teletalk: slow & steady

A low-volume forecast — small absolute numbers, modest but positive trend.

17 / THE DATA IS READY

Clean data. *Ready* to forecast.

576 tidy rows, zero nulls, two engineered features, and 51 flagged shocks — a model-ready foundation for the ensemble.

"Presentation 2 — EDA & Data Preprocessing for demand forecasting."

— CHRONOFORGE PULSE · DBI

Next · ARIMA + LSTM + XGBoost →

Then · 3 lot-sizing algorithms →

Questions & discussion ↗